

Hybrid Aerobic Development Plan

Base → Threshold → VO₂Max (12 Weeks)

CP: 260 W (W' 20 kJ) • Threshold Pace: 4:40/km

Update $\backslash currentCP$, $\backslash currentWprime$ and $\backslash currentCS$ — all watts and paces recompute. See Testing Protocol.

July 1, 2026

Phase 1 — Base: Sweet Spot & Volume (Weeks 1–4)

Sweet Spot, not pure Zone 2, is the primary quality tool this phase. Zone 2 isn't intrinsically superior for raising thresholds or VO₂max — at matched work, zone 3–5 training drives larger threshold/VO₂max gains per unit of effort (Inglis et al. 2024), and low-intensity volume is a complement to 2–3 weekly quality sessions, not a mandatory prerequisite before intervals (Matomäki 2025). Sweet spot is preferred over straight threshold here because it produces near-identical adaptations at meaningfully lower fatigue cost, letting SS Quality (WO2) and SS Long (WO4) coexist in the same week without compounding into an unrecoverable load. Progression is dosed as time-in-zone, not power: SS Quality adds a set/duration each week (2×20→2×25→3×20) while SS Long extends continuously (40→45→50 min) — staying under the ~2–2.5 h per-session ceiling and the 2×/week frequency guardrail past which sweet-spot fatigue starts to outpace its signal. Endurance runs and weekend Z2 rides progress the same way, duration before intensity, since extending duration is the correct lever until race-relevant duration is covered. **Before W1:** run the Critical Power / W' test and the threshold-pace test in the Testing Protocol section — don't reuse an old or estimated number; an overestimated CP turns "SS Quality" (90%) into threshold from day one.

W	WO1	WO2	WO3	WO4	WO5	WO6	WO7
1	Z1 Spin 90' @ 45–50% (117–130W)	SS Quality 2x20' @ 90% (234W) <i>4' rec, 85–95 rpm, baseline</i>	Tempo Intervals 3×12' @ 89–94% TP (4:58/km–5:15/km) <i>2' jog rec</i>	SS Long 1x40' @ 88–90% (229–234W) <i>continuous, TTE anchor</i>	Endurance Run 10 km @ 79–87% TP (5:22/km–5:54/km)	WO6 Long Z2 3 h @ 65–72% (169–187W)	WO7 Med Z2 2 h @ 62–70% (161–182W)
2	Z1 Spin 90' @ 45–50% (117–130W)	SS Quality 2x25' @ 90% (234W) <i>4' rec, +5' per interval</i>	Tempo Intervals 3×14' @ 89–94% TP (4:58/km–5:15/km) <i>2' jog rec</i>	SS Long 1x45' @ 88–90% (229–234W) <i>continuous, +5' vs W1</i>	Endurance Run 12 km @ 79–87% TP (5:22/km–5:54/km)	WO6 Long Z2 3.5 h @ 65–72% (169–187W)	WO7 Med Z2 2 h @ 62–70% (161–182W)
3	Z1 Spin 90' @ 45–50% (117–130W)	SS Quality 3x20' @ 90% (234W) <i>4' rec, +1 set vs W1</i>	Tempo Intervals 2×25' @ 89–94% TP (4:58/km–5:15/km) <i>2' jog rec</i>	SS Long 1x50' @ 88–90% (229–234W) <i>continuous, +5' vs W2</i>	Endurance Run 14 km @ 79–87% TP (5:22/km–5:54/km)	WO6 Long Z2 4 h @ 63–72% (164–187W)	WO7 Med Z2 2.5 h @ 62–70% (161–182W)
4	RECOVERY WEEK — Self-Managed						

Phase 2 — Threshold: CP Ceiling Development (Weeks 5–8)

Threshold succeeds sweet spot once SS Long's TTE growth signals the natural next step: W5's Threshold Entry (3×12' @ 95–100%) starts conservative and extends toward TTE (3×20' by W7), following the standard threshold progressive-overload rule — add time-in-zone per session, not more sets. The second weekly quality day (WO4) uses over/unders instead of a second steady-state threshold session: O/Us carry no unique physiological edge over steady-state work, but substituting one threshold day for O/Us adds race-specific accelerate-and-recover practice and keeps the week varied *without* adding load on top of the existing threshold day. Overs sit at 108% CP (~+20 W) rather than the commonly-misapplied 120% — that range crosses into VO₂max territory and defeats the point of a threshold session; unders sit at sweet spot (90% CP), matching the standard 1:4 bread-and-butter format (1 min over / 4 min under) and its documented progression (3×15→3×20→3×25 min). W8's recovery and CP/TP retest closes the phase and is the decision gate: if threshold TTE and CP are still moving, more threshold work is still the right tool; if they've stalled, that's the signal to move to VO₂ work (Phase 3) rather than add more threshold volume.

W	WO1	WO2	WO3	WO4	WO5	WO6	WO7
5	Z1 Spin 90' @ 45–50% (117–130W)	Threshold Entry 3x12' @ 95–100% (247–260W) <i>3' rec, start conservative</i>	Tempo Intervals 3×12' @ 89–94% TP (4:58/km–5:15/km) <i>2' jog rec</i>	O/U Short 3x15' (1' 108% (281W) / 4' 90% (234W)) <i>5' rec</i>	Endurance Run 14 km @ 79–87% TP (5:22/km–5:54/km)	WO6 Long Z2 3.5 h @ 65–72% (169–187W)	WO7 Med Z2 2 h @ 62–70% (161–182W)
6	Z1 Spin 90' @ 45–50% (117–130W)	Threshold Build 3x15' @ 95–100% (247–260W) <i>3' rec, even power</i>	Tempo Intervals 3×14' @ 89–94% TP (4:58/km–5:15/km) <i>2' jog rec</i>	O/U Medium 3x20' (1' 108% (281W) / 4' 90% (234W)) <i>5' rec</i>	Endurance Run 16 km @ 79–87% TP (5:22/km–5:54/km)	WO6 Long Z2 3.5 h @ 65–72% (169–187W)	WO7 Med Z2 2 h @ 62–70% (161–182W)
7	Z1 Spin 90' @ 45–50% (117–130W)	TTE Extension 3x20' @ 95–100% (247–260W) <i>5' rec, continuous preferred if legs</i>	Tempo Intervals 2×25' @ 89–94% TP (4:58/km–5:15/km) <i>2' jog rec</i>	O/U Long 3x25' (1' 108% (281W) / 4' 90% (234W)) <i>5' rec, peak threshold</i>	Endurance Run 18 km @ 79–87% TP (5:22/km–5:54/km)	WO6 Long Z2 4 h @ 63–72% (164–187W)	WO7 Med Z2 2.5 h @ 62–70% (161–182W)
8	RECOVERY WEEK — Self-Managed + CP/TP Retest						

Phase 3 — VO₂Max: Progressive Development (Weeks 9–12)

Two VO₂ sessions/week, not a concentrated block. A 5-session/week HIT block (Rønnestad et al. 2014) was considered and dropped: it compounds two variables — session frequency and per-rep intensity — that no cited study tested together, and it relies on a volume cutback and elite-level recovery capacity this plan doesn't assume. Progressive overload instead runs **within** the 2x/week format across the block: reps climb 4→5→6 and recovery trims 3→2.5→2 min week to week, while total work per session stays inside the 15–25 min ceiling — volume is not what's overloaded here, effort quality is. Per-interval execution follows the Empirical Cycling protocol: **no fixed power target** — start each rep 20–30% harder than you think you can hold, let power fade, keep pushing on perceived effort until breathing can no longer stay maxed. **Cadence 100–110 rpm** throughout — cadence, not power, is the primary lever for cardiac preload (stroke volume), which is the actual adaptation target. Recovery between reps: start shorter than instinct, add time only if the next rep can't reach max again. W9–11 carry full LIT volume alongside the 2 VO₂ days (easy run replaces tempo-interval running for this phase, protecting recovery around the two hard bike days). W12 is recovery; retest at the END of the week, not the start — VO₂max gains surface with a delay.

W	WO1	WO2	WO3	WO4	WO5	WO6	WO7
9	Z1 Spin 90' @ 45–50% (117–130W)	VO2 A 4x4' max mean / 100–110 rpm 3' rec, breathing maxed last 60–90s	Easy Run 12 km @ 65–77% TP (6:04/km–7:11/km)	WO4 Z2 1.5 h @ 60–68% (156–177W)	VO2 B 4x5' max mean / 100–110 rpm 3' rec, let power fade, hold effort	WO6 Long Z2 3.5 h @ 65–72% (169–187W)	WO7 Med Z2 2 h @ 62–70% (161–182W)
10	Z1 Spin 90' @ 45–50% (117–130W)	VO2 A 5x4' max mean / 100–110 rpm 2.5' rec, breathing maxed last 60–90s	Easy Run 14 km @ 65–77% TP (6:04/km–7:11/km)	WO4 Z2 1.5 h @ 60–68% (156–177W)	VO2 B 4x5' max mean / 100–110 rpm 2.5' rec, let power fade, hold effort	WO6 Long Z2 4 h @ 63–72% (164–187W)	WO7 Med Z2 2.5 h @ 62–70% (161–182W)
11	Z1 Spin 90' @ 45–50% (117–130W)	VO2 A 6x4' max mean / 100–110 rpm 2' rec, breathing maxed last 60–90s	Easy Run 16 km @ 65–77% TP (6:04/km–7:11/km)	WO4 Z2 1.5 h @ 60–68% (156–177W)	VO2 B 5x5' max mean / 100–110 rpm 2' rec, let power fade, hold effort	WO6 Long Z2 4 h @ 63–72% (164–187W)	WO7 Med Z2 2.5 h @ 62–70% (161–182W)
12	RECOVERY WEEK — Adaptation surfaces; CP/TP retest at END of week						

Appendix — Zone Reference

Cycling Zones at CP = 260 W

Zone	% CP	Watts
Z1 Recovery	<56%	<146W
Z2 Endurance	56–75%	146–195W
Z3 Tempo	76–87%	198–226W
Sweet Spot	88–94%	229–244W
Z4 Threshold	95–105%	247–273W
Z5 VO2 Max	106–120%	276–312W
Z6 Anaerobic	>121%	>315W

Running Zones at Threshold Pace = 4:40/km

Zone	% TP	Pace
Z1 Recovery	<77.5%	slower than 5:59/km
Z2 Endurance	78.5–87.7%	5:18/km–5:54/km
Z3 Tempo	88.7–94.3%	4:58/km–5:15/km
Z4 Threshold	95.3–100%	4:40/km–4:55/km
Z5a VO ₂ Max	101–103.4%	4:32/km–4:37/km
Z5b Anaerobic	104.4–111.5%	4:10/km–4:29/km
Z5c Neuromuscular	>112.5%	faster than 4:08/km

Testing Protocol — Critical Power (CP) & W'

This plan anchors all zones to **Critical Power (CP)**, not the classic “95% of 20-min” FTP shortcut. Head-to-head studies (Morgan 2019, Karsten 2021) show FTP and CP agree on average across a group but diverge by up to 15% for a given individual — enough to turn an intended Sweet Spot session into unintentional threshold work. CP is the field-testable proxy for the Maximum Metabolic Steady State (the boundary between sustainable and rapidly-fatiguing effort), derived from the power–duration relationship across multiple maximal efforts rather than a single 20-min time trial.

Protocol — 3 maximal efforts, all-out:

- **Efforts:** 3 min, 8 min, 12–13 min — all above CP (severe-intensity domain). Not shorter (risks hitting VO₂max/exhaustion before the effort is even representative) and not longer (glycogen, pacing, and motivation start dominating the result).
- **Recovery between efforts:** ≥30 min if done in one session, or spread across separate days (≥24 h) to remove fatigue carry-over between trials — prefer separate days if your schedule allows it.
- **Consistency:** same terrain across all three efforts (flat or a steady climb — whichever you train/race on most), same bike position, cadence held within ±5 rpm.
- **Pacing:** controlled fast start (~10% above goal power for the first 30 s, not all-out), even effort through the middle, a final 30 s kick if you have it left. Pick a route free of lights, traffic, and gradient changes.
- **Fitting CP and W' :** plot power vs. duration (or total work vs. duration) and fit all three standard models — 2-parameter hyperbolic, linear power-vs-1/time, linear work-vs-time. Take the **best individual fit**: whichever model gives the lowest combined error (SEE) for CP and W' on *your* data, not a fixed default model.
- **Quality check:** if CP’s SEE exceeds 5% or W' ’s SEE exceeds 10% across all three models, re-run whichever trial looks like it was paced poorly rather than trusting the fit.

Update \currentCP and \currentWprime with the result. Retest at the checkpoints already built into this plan: end of W8 (recovery week) and end of W12 (after the VO₂max adaptation window closes, not before). For running Threshold Pace, estimate the same way with two to three all-out time trials (e.g. 3 min + 9–13 min) on a flat, consistent route: $TP = (D_2 - D_1)/(T_2 - T_1)$.